

```
# Install required libraries
```

```
!pip install chromadb
```

```
!pip install sentence-transformers
```

```
!pip install langchain
```

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```

```
# Import required modules
```

```
import chromadb
```

```
from sentence_transformers import SentenceTransformer
```

```
# Create ChromaDB client
client = chromadb.Client()

# Create a collection (like a table)
collection = client.create_collection(name="rag_collection")
```

```
# Load pre-trained embedding model
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable h
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub
modules.json: 100% 349/349 [00:00<00:00, 7.37kB/s]

config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 6.09kB/s]

README.md: 10.5k/? [00:00<00:00, 271kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 637B/s]

config.json: 100% 612/612 [00:00<00:00, 13.9kB/s]

model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 209MB/s]

Loading weights: 100% 103/103 [00:00<00:00, 162.53it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+---
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect
tokenizer_config.json: 100% 350/350 [00:00<00:00, 22.1kB/s]

vocab.txt: 232k/? [00:00<00:00, 11.2MB/s]

tokenizer.json: 466k/? [00:00<00:00, 19.1MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 11.3kB/s]

config.json: 100% 190/190 [00:00<00:00, 17.1kB/s]
```

```
# Sample dataset (you can add more)
documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
    "Deep Learning uses neural networks.",
    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation.",
    "Neural networks are inspired by the human brain.",
    "Supervised learning uses labeled data.",
    "Unsupervised learning finds hidden patterns.",
    "Reinforcement learning learns through rewards.",
    "Transformers are used in modern NLP models."
]
```

```
# Unique IDs for each document
ids = [f"doc{i}" for i in range(len(documents))]
```

```
# Convert documents into vector embeddings
embeddings = model.encode(documents)
```

```
# Print shape of embeddings
print("Embedding shape:", embeddings.shape)
```

Embedding shape: (10, 384)

```
# Add data to ChromaDB collection
collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=ids
)

print("Documents stored successfully!")
```

Documents stored successfully!

```
# Define query
query = "What is Artificial Intelligence?"

# Convert query into embedding
query_embedding = model.encode([query])

# Perform similarity search
results = collection.query(
    query_embeddings=query_embedding,
    n_results=3
)

print(results)
```

```
{'ids': [['doc0', 'doc1', 'doc5']], 'embeddings': None, 'documents': [['Artificial Intelligence is
```

```
# Print retrieved documents
print("Top matching documents:\n")

for doc in results['documents'][0]:
    print("-", doc)
```

Top matching documents:

- Artificial Intelligence is the simulation of human intelligence.
- Machine Learning is a subset of AI.
- Neural networks are inspired by the human brain.

```
# Function for querying
def retrieve(query):
    query_embedding = model.encode([query])
    results = collection.query(
        query_embeddings=query_embedding,
        n_results=2
    )
    return results['documents'][0]

# Test function
print("\nCustom Query Result:")
print(retrieve("Explain machine learning"))
```

Custom Query Result:
['Machine Learning is a subset of AI.', 'Supervised learning uses labeled data.']

```
# Add more data later if needed
new_docs = [
    "AI is used in healthcare and finance.",
    "Chatbots use NLP to understand users."
]
```

```
J  
  
new_ids = ["doc_new1", "doc_new2"]  
new_embeddings = model.encode(new_docs)  
  
collection.add(  
    documents=new_docs,  
    embeddings=new_embeddings,  
    ids=new_ids  
)  
  
print("New documents added!")
```

New documents added!

Start coding or [generate](#) with AI.